

PAF 9272: Causal Analysis and Inference

Thomas Leavitt

Official Course Description

Causal Analysis and Inference (PAF 9272) is the second course in the research methods sequence. Upon completion of Data Collection and Description (PAF 9270), students can choose between this course and Data Analysis for Public Service (PAF 9271). PAF 9272 is meant for those interested in becoming analysts, researchers, or making quantitative data analysis an important element in their careers. It teaches students how to critically evaluate existing causal analyses of both qualitative and quantitative data and how to conduct statistical analyses to answer causal questions for domestic and international policy and practice. Compared to PAF 9271, PAF 9272 places greater emphasis on observational and experimental data from representative surveys and requires students to write programs (coding) to carry out statistical analyses using advanced statistical software such as Stata or R. The course provides a hands-on introduction to understanding causal evidence, covering logic models and mechanisms, case-oriented vs. variable-oriented approaches, correlation vs. causation, observational vs. (quasi) experimental data, treatment effect, confounding and omitted variable bias, complex survey sampling, generalizability, standard error, confidence interval estimation, hypothesis testing, statistical and practical significance, power analysis, multiple regression, and difference-in-differences estimation. Course sections will use applications tailored towards students' interests and concentrations (e.g., sections more populated with MIA students will have a greater international focus). (Students who took PAF 9170 or PAF 9172 cannot get credit for this course. They, and all other students looking for an advanced causal methods and statistics course, should consider taking PAF 9177.) Open to Austin W. Marxe School of Public and International Affairs MPA and MIA students; others with Marxe School permission.

Course Organization

This course begins by covering many of the topics introduced in PAF 9270 (Data Collection and Description), but does so with greater depth, rigor and accompanying statistical software.

The course then covers concepts and methods to draw conclusions about population quantities from randomly sampled data. After introducing different forms of random sampling, the course shows how to estimate a population average from sample data and test hypotheses about a population average or a difference in population averages.

The subsequent portion of the course covers concepts and methods to draw conclusions about policy impacts from comparisons of intervention and control groups, starting with randomized experiments, also known as A/B tests or RCTs (randomized controlled trials). The course then introduces observational studies, which depend on stronger assumptions than randomized experiments.

For all portions of the course, we will engage with real-world examples in a variety of policy domains. We will analyze and interpret existing data, with an emphasis on learning how to conduct analyses in R.

Course Learning Goals

1. Develop and describe causal theories, including mechanisms and alternative explanations of correlations, using qualitative and quantitative research and non-research sources.
2. Demonstrate understanding of causal research concepts, including causation, counterfactual, causal process tracing, generalizability, self-selection, control variables, omitted variable bias, experiments, random assignment, and training and validation datasets.
3. Analyze data with statistical software (such as Stata or R), interpret and write effectively about descriptive and inferential statistics, including hypothesis testing and power analysis.
4. Critically evaluate existing causal analyses of both qualitative and quantitative data, to draw valid and useful conclusions, including assessing generalizability, practical significance, and the quality of the causal evidence.
5. Programming (coding) with statistical software to accomplish data analysis tasks including cleaning data, merging complex datasets, creating new and transforming variables, visualizing data, and preparing tables and charts for professional reports and presentations.
6. Design and conduct research and analyses to estimate causal effects using experimental and observational data, with appropriate methods, such as multiple regression and difference-in-differences.

Course Assignments

Each student should complete 6 problem sets, each of which is due by 5:00 PM on the date listed in the course schedule below. Because I often review problem sets in class, I will not accept late assignments for credit.

Students should also complete a take-home final exam, due by 5:00 PM at the end of the semester as listed in the course schedule below, and all R assignments on the course's DataCamp site. Each DataCamp assignment expires at the beginning of class on the day it is due, and you cannot complete it afterward.

Grades

I will use the standard Baruch College grading scale. I reserve the right to curve the scale depending on the distribution of class scores at the end of the semester. No curve will lower any individual grades relative to what they would have been without the curve. Final course grades will weight the course assessments using the following percentages:

- 50% of your grade will be determined by 6 problem sets (8.333% each).
- 25% of your grade will be determined by the take-home final exam.
- 10% of your grade will be determined by the DataCamp exercises in R.
- 15% of your grade will be determined by course attendance and participation.

Course Readings

This course is ZTC (“Zero Textbook Cost”). All course readings will be available on the course’s Blackboard site. The course’s readings will draw primarily from the following books:

1. Freedman, D., R. Pisani, and R. Purves (2007). *Statistics* (4th ed.). New York, NY: W. W. Norton & Company.
2. Green, Donald P. (2022). *Social Science Experiments: A Hands-on Introduction*. New York, NY: Cambridge University Press.
3. Rosenbaum, Paul R. (2023). *Causal Inference*. Cambridge, MA: The MIT Press.

Course Standards

Electronics

Please feel free to use your laptops and tablets for class-related purposes. However, please do not use your cell phones or other electronic devices in class. If there are any students for whom cell phones or other electronics are important for course participation, please let me know. I will adjust the electronics policy for the entire class accordingly without singling out anyone.

Attendance, Participation and Collaboration

Attendance and active participation are extremely important to learning from this course. Except for the take-home exam, I strongly encourage collaboration with other students. However, students should write all take-home assignments individually, in their own words.

If you need to miss class, please do **not** contact me ahead of time. The implications of missing class for the attendance and participation components of your grade are the same regardless of whether you contact me beforehand. Please only contact me if you have to miss a class in which you are scheduled to present material.

Incomplete and Late Assignments

If you think that you will be unable to complete an assignment by its due date because of extenuating circumstances, please **do** let me know ahead of time. I may change the assignment's deadline for the entire class. For the final exam, if students turn it in late without prior consent from me, the exam's grade will be penalized by one-third of a letter grade per day (e.g., an A-exam that a student turns in one day late becomes a B+). For problem sets, because I typically review them in class, I will not accept late assignments for credit.

Use of Artificial Intelligence (AI) Tools

Because this course is fundamentally about quantitative reasoning and statistical computing, I see little sense in forbidding the very tools that are reshaping how such work is done. You are welcome—encouraged, even—to use large language models (ChatGPT, Claude, Copilot, and the like) to debug your R code, decipher cryptic error messages, brainstorm questions worth asking of the data, and draft the routine syntax you would otherwise type out by rote.

These tools have become genuinely capable: A single well-posed prompt can already carry you a surprising distance. Part of what I hope you take from this course, though, is a feel for how much further you can go once you pair prompting with real code, sensible data structures, and a little discipline in your workflow. Time and again we will see that a scripted, reproducible pipeline—one that can sweep through thousands of records, merge them with other sources, and keep a versioned log of every step—buys you a degree of scale, auditability, and reproducibility that ad hoc prompting cannot.

That latitude comes with one non-negotiable condition: You own everything you submit.

- **The output is yours.** “The model hallucinated” and “that is just what the chatbot gave me” are never valid defenses. If your code quietly returns the wrong answer, or your paper cites a study that was never written, the error is yours and no one else's. Treat these tools as you would an eager assistant who is frequently confident and occasionally completely wrong: Verify everything they produce before your name goes on it.
- **You must be able to explain it.** You may use AI to help write code, but you are answerable for every line you hand in. If I ask you in class or in office hours to walk me through a particular function or loop in your work and you cannot say how it behaves or why it is there, that portion of the assignment may receive no credit. More important still, I want you to ask at every step what could be wrong with a given result—what sanity check, cross-tabulation, or quick plot would tell you whether you are truly on the right track.
- **Think before you prompt.** These tools are excellent at carrying out a task you have already framed and poor at supplying the insight that framed it. Do not surrender your judgment to them; use them to move faster on questions that are already your own, not to do your thinking for you.

Finally, disclosure. Any assignment on which you relied on AI tools must carry a brief disclosure statement. As journals ask of their authors, it should (i) name the specific tool(s) and the version or model you used (for example, the product name together with its model or version identifier), (ii) state what you used each one for—which tasks, and which parts of the work—and (iii) confirm that you reviewed and verified the output and take full responsibility for

it. As in those venues, an AI tool is never a co-author and never answerable for the work; that responsibility is yours alone.

The course's academic-integrity standards apply to AI-assisted work exactly as they do to everything else, and I treat undisclosed use of AI as a breach of them. These tools also routinely misattribute or invent their sources, so catching and correcting such errors before you submit is part of the same responsibility.

Academic Integrity

I will not tolerate violations of academic integrity under any circumstances. Please familiarize yourself with [CUNY's Academic Integrity Policy](#). Additional resources to consult are Graff and Birkenstein (2021), Lipson (2008) and Neville (2016). If you are having difficulty with a written assignment, do not attempt to present another author's work as your own; please come talk to me instead.

Academic Student Support

Baruch's Marx School offers academic student support services for both writing and quantitative skills. I highly recommend looking into the Quantitative Student Support department's tutoring services and workshops. More information about these tutoring services and workshops is available [here](#).

Software and computing

This course will use the R statistical programming language. To download R, please go to <https://cloud.r-project.org> and then follow the relevant instructions for installation.

Once you have successfully installed R, please make sure to download and install RStudio, which is an integrated development environment for R. You can download and install RStudio at <https://posit.co/download/rstudio-desktop/>.

You should install both R and RStudio on a personal laptop that you bring to every class. If you do not have access to a personal laptop or other technology, Baruch College's Technology Loan Program may be able to help. You can find more information about this Technology Loan Program [here](#). You may also want to consider use of [Posit Cloud](#), which enables you to use R and RStudio directly in an internet browser without downloading them to a personal device.

A valuable resource for learning R is Wickham et al. (2023), which is available for free at <https://r4ds.hadley.nz/>.

In addition, to facilitate students' learning R, all students should complete all assignments on the course's DataCamp site.

AI tools can be particularly helpful for coding with statistical software, especially R. For help with such coding, I recommend ChatGPT, which you can access [here](#). Using ChatGPT requires an account, but a free plan is available.

Course Schedule and Readings

Baruch College's academic calendar is available via the Office of the Registrar [here](#).

Session 1: Introduction to course and the R programming language

- ★ Problem set 1 distributed

Session 2: Descriptive statistics — summarizing empirical distributions

- ★ Problem set 2 distributed
- ★ Problem set 1 due
- ★ DataCamp Assignment due
 - Introduction to R: Intro to basics

Required readings

Theory: Freedman et al. (2007, pp. 57–96)

Session 3: Descriptive statistics — summarizing relationships between variables

- ★ DataCamp Assignment due
 - Introduction to R: Vectors

Required readings

Theory: Freedman et al. (2007, pp. 119–157)

Session 4: Random sampling

- ★ Problem set 3 distributed
- ★ Problem set 2 due
- ★ DataCamp Assignment due
 - Introduction to R: Matrices

Required readings

Theory: Freedman et al. (2007, pp. 333–354)

Application(s): Menachemi et al. (2020, 2021) and Kishore et al. (2018)

Session 5: Estimation and inference of population quantities I

- ★ DataCamp Assignment due
 - Introduction to R: Factors

Required readings

Theory: Freedman et al. (2007, pp. 355–376 and pp. 409–437)

Application(s): Menachemi et al. (2020, 2021) and Kishore et al. (2018)

Session 6: Estimation and inference of population quantities II

- ★ [DataCamp Assignment due](#)

- Introduction to R: Data frames

Required readings

Theory: Freedman et al. (2007, pp. 475–500 and pp. 545–576)

Application(s): Menachemi et al. (2020, 2021) and Kishore et al. (2018)

Session 7: Causality and research design

- ★ Problem set 4 distributed

- ★ [Problem set 3 due](#)

- ★ [DataCamp Assignment due](#)

- Introduction to R: Lists

Required readings

Theory: Green (2022, pp. 1–22)

Session 8: Introduction to randomized experiments

- ★ [DataCamp Assignment due](#)

- Intermediate R: Conditionals and Control Flow

Required readings

Theory: Rosenbaum (2023, pp. 1–45)

Application(s): Sherman et al. (1995)

Recommended readings

Green (2022, pp. 42–66)

Session 9: Estimation and inference of causal effects in randomized experiments I

- ★ Problem set 5 distributed

- ★ [Problem set 4 due](#)

Required readings

Theory: Gerber and Green (2012, pp. 21–50)

Application(s): Sherman et al. (1995)

Session 10: Estimation and inference of causal effects in randomized experiments II

- ★ [DataCamp Assignment due](#)

- Introduction to the Tidyverse: Data wrangling

Required readings

Theory: Gerber and Green (2012, pp. 51–66)

Application(s): Sherman et al. (1995)

Session 11: Introduction to observational designs

- ★ Problem set 6 distributed

- ★ [Problem set 5 due](#)

- ★ [DataCamp Assignment due](#)

- Introduction to the Tidyverse: Data visualization

Required readings

Theory: Rosenbaum (2023, pp. 47–65)

Application(s): Silber et al. (2022)

Session 12: Stratification and adjustment for measured covariates in observational designs

Required readings

Theory: Rosenbaum (2023, pp. 67–84)

Application(s): Silber et al. (2022)

Session 13: Pre-post and controlled pre-post designs

- ★ [Problem set 6 due](#)

- ★ [DataCamp Assignment due](#)

- Introduction to the Tidyverse: Grouping and summarizing

Required readings

Theory: Shadish et al. (2002, pp. 103–147)

Application(s): MTA's Fare-Free Bus Pilot Evaluation (MTA, 2024; BronxNet, 2024)

Session 14: Conclusion and pre-final exam review

- ★ Final exam due (at the end of the semester)
- ★ DataCamp Assignment due
 - Introduction to the Tidyverse: Types of visualizations

Baruch Policies and Resources

Equal Opportunity and Non-Discrimination

You can submit a report of discrimination and/or retaliation via CUNY's centralized reporting platform [here](#). For more information about CUNY's Policy on Equal Opportunity and Non-Discrimination ("EO Policy"), please access it [here](#). You can also access CUNY's revised Policy on Sexual Misconduct [here](#).

Student Disability Services

Baruch College's Student Disability Services provides accommodations to students with disabilities to promote equal access to Baruch's programs and services. To request accommodations, please do so as soon as possible by directly contacting Student Disability Services [here](#).

Campus Intervention Team (CIT)

The CIT provides a support system for students in crisis. Any member of Baruch's community (faculty, students, staff) can report a concern about a student to CIT. **All reports are confidential.** You can learn more about CIT and the services it offers [here](#).

To report a concern related to concerning behavior, student conduct, academic integrity or student grievance, please use Baruch College's Portal for Incident Reporting available [here](#).

Writing Support

The Schwartz Communication Institute

Writing support services for graduate students in the Marx school are available through a partnership with the [Bernard L. Schwartz Communication Institute](#) at the Center for Teaching and Learning. Please reach out to the Assistant Director for Writing in Public and International Affairs for more information on one-to-one writing consultations. You can also visit the website for writing resources and workshops designed to support graduate students at Marx. You can learn more [here](#).

The Baruch Writing Center

The Writing Center offers free, professional writing support for all undergraduate and graduate students at Baruch, through one-to-one consultations, workshops, peer review groups, written feedback, online resources, and a journal of outstanding student writing. The Writing Center supports faculty through classroom visits, in-class workshops, referral forms, and workshop lesson plans, and it is always available for conversations about teaching and writing. More information is available [here](#), by calling 646-312-4012 or by emailing the center at writing.center@baruch.cuny.edu.

Academic Advisement

The Marx school provides a range of academic advisement services to ensure the successful completion of students' degree programs. For undergraduate advisement (Bachelor of Science

in Public Affairs – BSPA), you can write an e-mail to mbspia.bspa@baruch.cuny.edu. For graduate advisement (Master of Public Administration, Executive MPA, Master of International Affairs, and Master of Science in Education-HEA or any of Marxé's certificate programs), you can write an e-mail to mbspia.advisement@baruch.cuny.edu.

You can also schedule an appointment with an academic advisor via Baruch College's Navigate platform available [here](#).

Career Services and Resources

For personalized career guidance, please make an appointment by logging in to [Marxé Career-Connect](#). For general inquiries, you can write an e-mail to mbspia.careerservices@baruch.cuny.edu. More information about career services is available [here](#); a variety of career resources, including important links, documents and opportunities, is available [here](#).

References

- BronxNet (2024, April). Today's Verdict: Push to expand bus pilot program. <https://www.youtube.com/watch?v=oUbbA8GBAw0>.
- Freedman, D., R. Pisani, and R. Purves (2007). *Statistics* (4th ed.). New York, NY: W. W. Norton & Company.
- Gerber, A. S. and D. P. Green (2012). *Field Experiments: Design, Analysis, and Interpretation*. New York, NY: W.W. Norton.
- Graff, G. and C. Birkenstein (2021). *They Say / I Say* (5th ed.). New York, NY: W. W. Norton & Company.
- Green, D. P. (2022). *Social Science Experiments: A Hands-on Introduction*. New York, NY: Cambridge University Press.
- Kishore, N., D. Marqués, A. Mahmud, M. V. Kiang, I. Rodriguez, A. Fuller, P. Ebner, C. Sorensen, F. Racy, J. Lemery, L. Maas, J. Leaning, R. A. Irizarry, S. Balsari, and C. O. Buckee (2018). Mortality in Puerto Rico after Hurricane Maria. *The New England Journal of Medicine* 379(2), 162–170.
- Lipson, C. (2008). *Doing Honest Work in College: How to Prepare Citations, Avoid Plagiarism, and Achieve Real Academic Success* (2nd ed.). Chicago, IL: University of Chicago Press.
- Menachemi, N., B. E. Dixon, K. K. Wools-Kaloustian, C. T. Yiannoutsos, and P. K. Halverson (2021). How many SARS-CoV-2-Infected people require hospitalization? Using random sample testing to better inform preparedness efforts. *Journal of Public Health Management and Practice* 27(3), 246–250.
- Menachemi, N., C. T. Yiannoutsos, B. E. Dixon, T. J. Duszynski, W. F. Fadel, K. K. Wools-Kaloustian, N. U. Needleman, K. Box, V. Caine, C. Norwood, L. Weaver, and P. K. Halverson (2020). Population point prevalence of SARS-CoV-2 infection based on a statewide random sample – Indiana, April 25–29, 2020. *The Morbidity and Mortality Weekly Report (MMWR)* 69, 960–964.
- MTA (2024, July). Fare-free bus pilot evaluation. <https://mta.info/document/147096>.
- Neville, C. (2016). *The Complete Guide to Referencing and Avoiding Plagiarism* (3rd ed.). Buckingham, UK: Open University Press.
- Rosenbaum, P. R. (2023). *Causal Inference*. The MIT Press Essential Knowledge Series. Cambridge, MA: The MIT Press.
- Shadish, W. R., T. D. Cook, and D. T. Campbell (2002). *Experimental and Quasi-Experimental Designs for Generalized Causal Inference*. Boston, MA: Houghton Mifflin Company.
- Sherman, L. W., D. P. Rogan, T. Edwards, R. Whipple, D. Shreve, D. Witcher, W. Trimble, The Street Narcotics Unit, R. Velke, M. Blumberg, A. Beatty, and C. A. Bridgeforth (1995). Deterrent effects of police raids on crack houses: A randomized, controlled experiment. *Justice Quarterly* 12(4), 755–781.

Silber, J. H., P. R. Rosenbaum, J. G. Reiter, A. S. Hill, S. Jain, D. A. Wolk, D. Small, S. Hashemi, B. A. Niknam, M. D. Neuman, L. A. Fleisher, and R. Eckenhoff (2022). Alzheimer’s dementia after exposure to anesthesia and surgery in the elderly: A matched natural experiment using appendicitis. *Annals of Surgery* 276(5), e377–e385.

Wickham, H., M. Çetinkaya Rundel, and G. Grolemund (2023). *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data* (2nd ed.). Sebastopol, CA: O’Reilly Media.